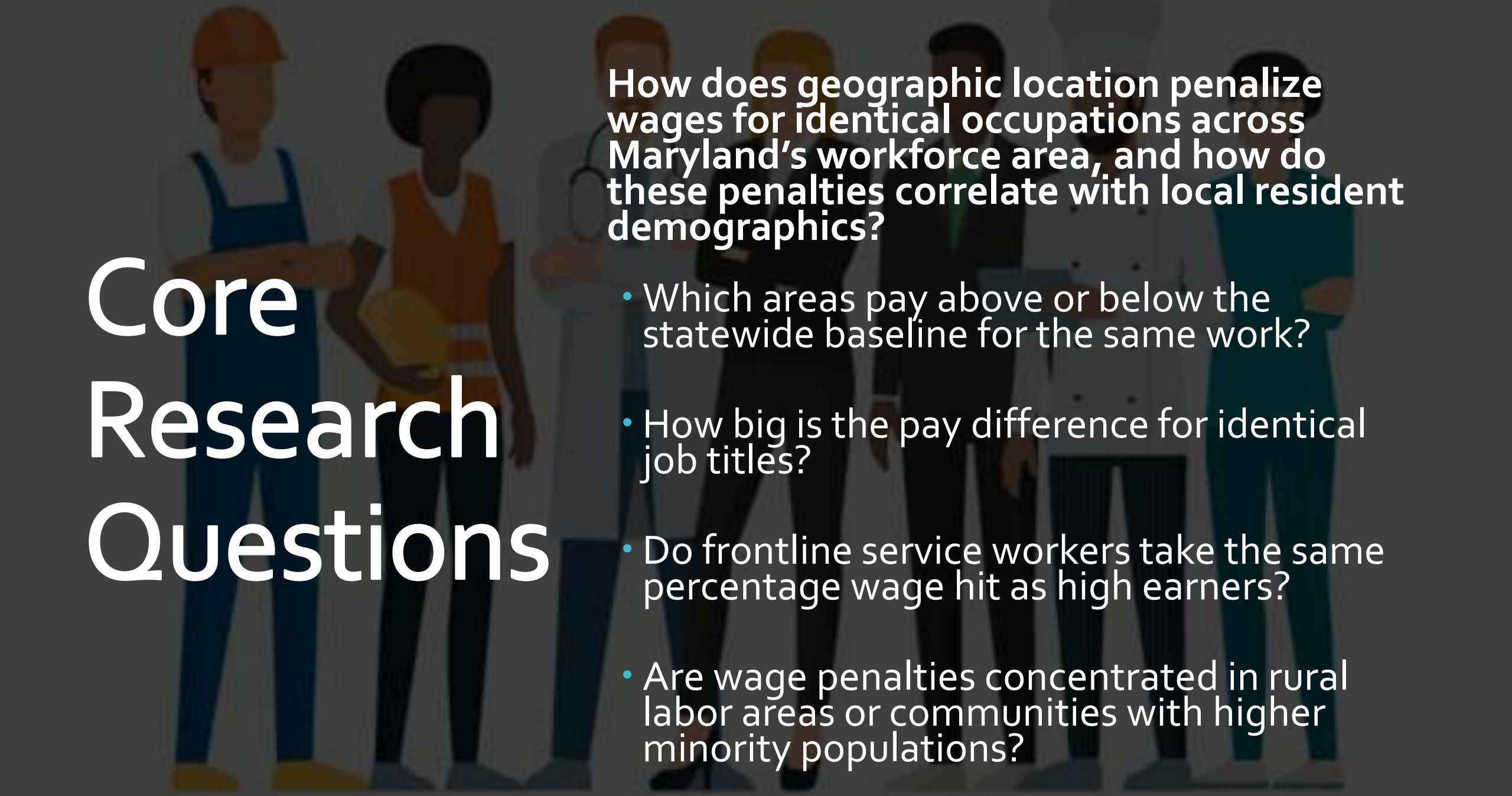
# Analyzing Occupational Wage Gaps Across Maryland

Rin Hwang

DATA 205



# Core Research Questions

**How does geographic location penalize wages for identical occupations across Maryland's workforce area, and how do these penalties correlate with local resident demographics?**

- Which areas pay above or below the statewide baseline for the same work?
- How big is the pay difference for identical job titles?
- Do frontline service workers take the same percentage wage hit as high earners?
- Are wage penalties concentrated in rural labor areas or communities with higher minority populations?

# Datasets, Analysis and Modeling

- Datasets (2):
  - "2025 Occupational Employment & Wage Statistics"
    - Maryland Department of Labor, Division of Workforce Development and Adult Learning
  - "Annual Estimates of the Resident Population by Demographic Characteristics for Maryland Jurisdictions"
    - Maryland Department of Planning, State Data and Analysis Center
- Multiple Linear Regression
  - Using location dummy variables to get a numerical penalty
- Two-Way ANOVA
  - Testing whether wage gaps vary significantly across occupations and locations
- Logistic Regression
  - Are frontline workers being disproportionally affected by wage gaps?

# Planned Visuals

- Geographic Map of Wage Penalties
  - Maryland state map to show severity of wage penalties
  - Colored scaled shaded counties (choropleth), the darker the steeper the wage penalty is
- Correlations of Demographics Scatterplots
  - Equity patterns across communities
  - Whether communities with higher minority populations face steeper wage penalties
- Boxplots of Penalties
  - Wage penalty distributions grouped across occupations
  - Is every occupation getting hit with penalties?
  - Are there certain occupations that are more stable?

# Capstone Project: Wage Gaps in Maryland

**Project Author:** Rin Hwang
**Final Presentation Date:** December 15, 2026

## TIMELINE



# Milestones and Potential Risks

- Project Progression
  - Preparing both datasets by cleaning and wrangling
  - Start doing modeling: Run regression and ANOVA models to get baseline regional penalties.
  - Merge demographic data, run equity models, and finalize visuals
- Risks and Solutions
  - Missing data in rural counties
    - Solution: Only comparing specific jobs with enough reported data and using broad career fields whenever a local sample gets too small
  - Work location ≠ Residence
    - People don’t always live in the same county where they work.
    - Solution: Grouping the data into the 13 local workforce areas and focusing only on what local employers pay

# Sources

Datasets:

Maryland Department of Labor, Division of Workforce Development and Adult Learning. (2025). Maryland occupational employment and wage statistics (OEWS). Maryland Department of Labor. https://labor.maryland.gov/lmi/wages/

Maryland Department of Planning, State Data and Analysis Center. (2025). Annual Estimates of the Resident Population by Demographic Characteristics for Maryland Jurisdictions. State of Maryland. https://planning.maryland.gov/MSDC/SiteAssets/Census_Estimates/Age-Race-Sex/Census_EST_ARS.aspx

Images:

Slide 1 Flag of Maryland: https://en.wikipedia.org/wiki/Flag_of_Maryland
Slide 2 Workers stock image: https://www.istockphoto.com/illustrations/different-occupations
Slide 3 Graphs stock image: https://www.vecteezy.com/free-photos/statistical-analysis
Slide 4 Graphs stock image: https://stock.adobe.com/search?k=graphs
Slide 5 Gantt Chart: Created on Canva using template
Slide 6 Computer stock image: https://www.istockphoto.com/photos/market-research